Time limits. The 1,800-second budget per instance comprises 1,680 seconds for the exact search and up to 120 seconds for a fallback heuristic. The reported times include the actual heuristic runtime, which may be much shorter than 120 seconds; thus, runs without a proof of optimality can finish in slightly more than 1,680 seconds.

1. Detailed Results of Uniform-Weight Instances on S_A under WTFRP Setting

We first report the results of Ha26 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 1 Detailed Results of Ha26 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	1081.0	0.00	12.33

1.1. Results on the instances with i execution mode

Table 2 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	1062.0	0.00	0.92
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	1079.0	0.00	10.87
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	6790.0	0.00	29.17
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	6688.0	0.00	14.15
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 3 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1043.0	0.00	9.65	60-10-i-1	1042.0	0.00	0.81
60-10-i-2	1062.0	0.00	8.51	60-10-i-2	1057.0	0.00	0.98	60-10-i-2	1056.0	0.00	0.81
60-10-i-3	1077.0	0.00	0.87	60-10-i-3	1077.0	0.00	9.37	60-10-i-3	1077.0	0.00	10.61
60-10-i-4	-	-	-	60-10-i-4	1063.0	0.00	0.96	60-10-i-4	1063.0	0.00	0.93
60-10-i-5	1067.0	0.00	0.94	60-10-i-5	1067.0	0.00	1.68	60-10-i-5	1066.0	0.00	0.86
60-20-i-1	2151.0	0.00	0.85	60-20-i-1	2150.0	0.00	1.45	60-20-i-1	2150.0	0.00	1.29
60-20-i-2	2133.0	0.00	9.29	60-20-i-2	2130.0	0.00	8.28	60-20-i-2	2129.0	0.00	7.77
60-20-i-3	2028.0	0.00	8.06	60-20-i-3	2026.0	0.00	7.11	60-20-i-3	2025.0	0.00	0.76
60-20-i-4	-	-	-	60-20-i-4	2028.0	0.00	0.80	60-20-i-4	2026.0	0.00	0.74
60-20-i-5	-	-	-	60-20-i-5	2150.0	0.00	1.25	60-20-i-5	2150.0	0.00	0.64
60-30-i-1	3134.0	0.00	4.67	60-30-i-1	3129.0	0.00	0.75	60-30-i-1	3127.0	0.00	0.65
60-30-i-2	-	-	-	60-30-i-2	3210.0	0.00	8.42	60-30-i-2	3204.0	0.00	7.18
60-30-i-3	-	-	-	60-30-i-3	3184.0	0.00	1.53	60-30-i-3	3184.0	0.00	1.26
60-30-i-4	-	-	-	60-30-i-4	3182.0	0.00	7.77	60-30-i-4	3181.0	0.00	2.06
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	3191.0	0.00	1.29
90-10-i-1	-	-	-	90-10-i-1	1636.0	0.00	1.86	90-10-i-1	1636.0	0.00	7.39
90-10-i-2	-	-	-	90-10-i-2	1676.0	0.00	0.74	90-10-i-2	1676.0	0.00	0.76
90-10-i-3	1625.0	0.00	0.78	90-10-i-3	1625.0	0.00	9.62	90-10-i-3	1624.0	0.00	0.79
90-10-i-4	-	-	-	90-10-i-4	1650.0	0.00	8.71	90-10-i-4	1649.0	0.00	0.75
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	1644.0	0.00	9.46
90-20-i-1	-	-	-	90-20-i-1	3307.0	0.00	1.32	90-20-i-1	3307.0	0.00	1.27
90-20-i-2	-	-	-	90-20-i-2	3328.0	0.00	8.80	90-20-i-2	3327.0	0.00	0.69
90-20-i-3	3331.0	0.00	9.52	90-20-i-3	3328.0	0.00	7.64	90-20-i-3	3327.0	0.00	0.64
90-20-i-4	3321.0	0.00	0.96	90-20-i-4	3316.0	0.00	4.09	90-20-i-4	3315.0	0.00	1.29
90-20-i-5	-	-	-	90-20-i-5	3288.0	0.00	0.80	90-20-i-5	3286.0	0.00	4.16
90-30-i-1	4952.0	0.00	3.17	90-30-i-1	4948.0	0.00	1.30	90-30-i-1	4948.0	0.00	0.72
90-30-i-2	-	-	-	90-30-i-2	4837.0	0.00	5.23	90-30-i-2	4835.0	0.00	1.50
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	5012.0	0.00	0.81
90-30-i-4	4721.0	0.00	9.72	90-30-i-4	4720.0	0.00	1.85	90-30-i-4	4720.0	0.00	0.69
90-30-i-5	4930.0	0.00	2.06	90-30-i-5	4928.0	0.00	0.94	90-30-i-5	4928.0	0.00	1.37
120-10-i-1	2209.0	0.00	0.77	120-10-i-1	2209.0	0.00	9.64	120-10-i-1	2209.0	0.00	0.72
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	2216.0	0.00	0.74
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	2223.0	0.00	0.73
120-10-i-4	-	-	-	120-10-i-4	2176.0	0.00	0.74	120-10-i-4	2176.0	0.00	0.74
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	2252.0	0.00	8.83
120-20-i-1	4583.0	0.00	0.81	120-20-i-1	4583.0	0.00	1.24	120-20-i-1	4583.0	0.00	1.26
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	4541.0	0.00	0.78
120-20-i-3	4525.0	0.00	10.38	120-20-i-3	4520.0	0.00	1.61	120-20-i-3	4520.0	0.00	0.72
120-20-i-4	4432.0	0.00	3.80	120-20-i-4	4431.0	0.00	4.91	120-20-i-4	4431.0	0.00	3.27
120-20-i-5	4485.0	0.00	11.40	120-20-i-5	4485.0	0.00	8.24	120-20-i-5	4485.0	0.00	9.01
120-30-i-1	6721.0	0.00	46.49	120-30-i-1	6717.0	0.00	5.79	120-30-i-1	6717.0	0.00	9.56
120-30-i-2	6784.0	0.00	2.87	120-30-i-2	6784.0	0.00	0.96	120-30-i-2	6784.0	0.00	1.54
120-30-i-3	6685.0	0.00	1.64	120-30-i-3	6685.0	0.00	3.92	120-30-i-3	6685.0	0.00	1.43
120-30-i-4	6702.0	0.00	2.31	120-30-i-4	6702.0	0.00	8.61	120-30-i-4	6700.0	0.00	1.42
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	6635.0	0.00	2.01

1.2. Results on the instances with sf execution mode

Table 4 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1044.0	0.00	7.63	60-10-sf-1	1035.0	0.00	3.34	60-10-sf-1	1033.0	0.00	2.52
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1051.0	0.00	8.11
60-10-sf-3	1074.0	0.00	9.12	60-10-sf-3	1070.0	0.00	8.65	60-10-sf-3	1067.0	0.00	5.60
60-10-sf-4	1062.0	0.00	1.79	60-10-sf-4	1059.0	0.00	11.38	60-10-sf-4	1053.0	0.00	0.94
60-10-sf-5	1061.0	0.00	10.58	60-10-sf-5	1059.0	0.00	1.83	60-10-sf-5	1057.0	0.00	3.39
60-20-sf-1	-	-	-	60-20-sf-1	2142.0	0.00	6.27	60-20-sf-1	2136.0	0.00	12.08
60-20-sf-2	-	-	-	60-20-sf-2	2129.0	0.00	11.12	60-20-sf-2	2122.0	0.00	16.32
60-20-sf-3	-	-	-	60-20-sf-3	2026.0	0.00	4.16	60-20-sf-3	2022.0	0.00	10.63
60-20-sf-4	-	-	-	60-20-sf-4	2017.0	0.00	9.95	60-20-sf-4	2010.0	0.00	10.53
60-20-sf-5	-	-	-	60-20-sf-5	2146.0	0.00	11.99	60-20-sf-5	2136.0	0.00	8.81
60-30-sf-1	-	-	-	60-30-sf-1	3106.0	0.00	15.20	60-30-sf-1	3099.0	0.00	40.44
60-30-sf-2	-	-	-	60-30-sf-2	3201.0	0.00	1151.58	60-30-sf-2	3182.0	0.00	529.18
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	3168.0	0.00	29.81
60-30-sf-4	-	-	-	60-30-sf-4	3177.0	0.00	192.72	60-30-sf-4	3162.0	0.00	235.77
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	3171.0	0.00	20.31
90-10-sf-1	-	-	-	90-10-sf-1	1625.0	0.00	11.79	90-10-sf-1	1621.0	0.00	8.05
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	1663.0	0.00	1.83
90-10-sf-3	-	-	-	90-10-sf-3	1613.0	0.00	1.65	90-10-sf-3	1611.0	0.00	0.85
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	1639.0	0.00	9.73	90-10-sf-5	1634.0	0.00	9.34	90-10-sf-5	1631.0	0.00	6.33
90-20-sf-1	-	-	-	90-20-sf-1	3288.0	0.00	16.05	90-20-sf-1	3279.0	0.00	11.00
90-20-sf-2	-	-	-	90-20-sf-2	3302.0	0.00	63.63	90-20-sf-2	3293.0	0.00	19.87
90-20-sf-3	-	-	-	90-20-sf-3	3305.0	0.00	13.36	90-20-sf-3	3296.0	0.00	13.10
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	3299.0	0.00	5.46
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	3259.0	0.00	5.93
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	4914.0	0.00	699.78
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	4813.0	0.00	52.37
90-30-sf-3	-	-	-	90-30-sf-3	4988.0	0.00	1325.76	90-30-sf-3	4976.0	0.00	101.58
90-30-sf-4	-	-	-	90-30-sf-4	4702.0	0.00	153.47	90-30-sf-4	4681.0	0.00	896.55
90-30-sf-5	-	-	-	90-30-sf-5	4899.0	0.00	123.86	90-30-sf-5	4887.0	0.00	12.55
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	2195.0	0.00	8.36
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	2205.0	0.00	9.31
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	2207.0	0.00	10.26
120-10-sf-4	2166.0	0.00	6.34	120-10-sf-4	2155.0	0.00	8.10	120-10-sf-4	2155.0	0.00	8.46
120-10-sf-5	-	-	-	120-10-sf-5	2244.0	0.00	11.69	120-10-sf-5	2236.0	0.00	13.09
120-20-sf-1	4554.0	0.00	927.66	120-20-sf-1	4541.0	0.00	24.92	120-20-sf-1	4536.0	0.00	6.31
120-20-sf-2	-	-	-	120-20-sf-2	4514.0	0.00	21.85	120-20-sf-2	4503.0	0.00	12.61
120-20-sf-3	-	-	-	120-20-sf-3	4501.0	0.00	1707.85	120-20-sf-3	4475.0	0.00	370.82
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	6683.0	2.39	1684.74	120-30-sf-1	6659.0	0.00	18.71
120-30-sf-2	-	-	-	120-30-sf-2	6750.0	0.00	1689.54	120-30-sf-2	6732.0	0.00	177.16
120-30-sf-3	-	-	-	120-30-sf-3	6643.0	0.00	174.26	120-30-sf-3	6635.0	0.00	241.51
120-30-sf-4	-	-	-	120-30-sf-4	6665.0	0.00	614.05	120-30-sf-4	6649.0	0.00	1684.20
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 5 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1031.0	0.00	0.96	60-10-sf-1	1031.0	0.00	0.96	60-10-sf-1	1031.0	0.00	1.58
60-10-sf-2	1049.0	0.00	8.29	60-10-sf-2	1049.0	0.00	8.32	60-10-sf-2	1048.0	0.00	10.43
60-10-sf-3	1066.0	0.00	2.45	60-10-sf-3	1063.0	0.00	11.89	60-10-sf-3	1063.0	0.00	11.06
60-10-sf-4	1052.0	0.00	1.77	60-10-sf-4	1052.0	0.00	1.48	60-10-sf-4	1051.0	0.00	0.86
60-10-sf-5	1055.0	0.00	1.84	60-10-sf-5	1054.0	0.00	1.82	60-10-sf-5	1053.0	0.00	0.86
60-20-sf-1	2132.0	0.00	10.08	60-20-sf-1	2132.0	0.00	1.09	60-20-sf-1	2132.0	0.00	1.50
60-20-sf-2	2115.0	0.00	10.26	60-20-sf-2	2109.0	0.00	1.03	60-20-sf-2	2108.0	0.00	9.90
60-20-sf-3	2016.0	0.00	11.46	60-20-sf-3	2016.0	0.00	9.51	60-20-sf-3	2014.0	0.00	0.81
60-20-sf-4	2007.0	0.00	6.32	60-20-sf-4	2006.0	0.00	8.68	60-20-sf-4	2005.0	0.00	2.17
60-20-sf-5	2134.0	0.00	5.13	60-20-sf-5	2133.0	0.00	8.03	60-20-sf-5	2132.0	0.00	0.73
60-30-sf-1	3095.0	0.00	14.76	60-30-sf-1	3092.0	0.00	3.42	60-30-sf-1	3092.0	0.00	5.12
60-30-sf-2	3169.0	0.00	57.82	60-30-sf-2	3166.0	0.00	13.98	60-30-sf-2	3165.0	0.00	0.78
60-30-sf-3	3159.0	0.00	12.29	60-30-sf-3	3155.0	0.00	1.99	60-30-sf-3	3155.0	0.00	2.04
60-30-sf-4	3155.0	0.00	21.18	60-30-sf-4	3151.0	0.00	11.12	60-30-sf-4	3150.0	0.00	8.64
60-30-sf-5	3163.0	0.00	22.93	60-30-sf-5	3161.0	0.00	4.59	60-30-sf-5	3160.0	0.00	1.41
90-10-sf-1	1621.0	0.00	7.69	90-10-sf-1	1617.0	0.00	8.96	90-10-sf-1	1617.0	0.00	2.31
90-10-sf-2	1659.0	0.00	5.32	90-10-sf-2	1658.0	0.00	0.76	90-10-sf-2	1658.0	0.00	8.34
90-10-sf-3	1610.0	0.00	0.85	90-10-sf-3	1610.0	0.00	0.96	90-10-sf-3	1610.0	0.00	0.85
90-10-sf-4	1638.0	0.00	9.23	90-10-sf-4	1635.0	0.00	1.51	90-10-sf-4	1635.0	0.00	9.17
90-10-sf-5	1629.0	0.00	5.74	90-10-sf-5	1629.0	0.00	2.91	90-10-sf-5	1629.0	0.00	1.56
90-20-sf-1	3275.0	0.00	5.00	90-20-sf-1	3273.0	0.00	1.49	90-20-sf-1	3273.0	0.00	7.96
90-20-sf-2	3290.0	0.00	9.85	90-20-sf-2	3290.0	0.00	3.16	90-20-sf-2	3289.0	0.00	1.38
90-20-sf-3	3295.0	0.00	5.53	90-20-sf-3	3295.0	0.00	6.21	90-20-sf-3	3295.0	0.00	4.66
90-20-sf-4	3291.0	0.00	12.25	90-20-sf-4	3286.0	0.00	2.83	90-20-sf-4	3285.0	0.00	0.77
90-20-sf-5	3254.0	0.00	9.80	90-20-sf-5	3250.0	0.00	1.53	90-20-sf-5	3249.0	0.00	0.85
90-30-sf-1	4902.0	2.04	1682.95	90-30-sf-1	4890.0	0.00	12.17	90-30-sf-1	4888.0	0.00	2.69
90-30-sf-2	4798.0	0.00	9.23	90-30-sf-2	4794.0	0.00	3.91	90-30-sf-2	4790.0	0.00	0.95
90-30-sf-3	4970.0	0.00	22.42	90-30-sf-3	4968.0	0.00	28.74	90-30-sf-3	4966.0	0.00	2.38
90-30-sf-4	4674.0	0.00	7.86	90-30-sf-4	4673.0	0.00	4.35	90-30-sf-4	4672.0	0.00	0.88
90-30-sf-5	4885.0	0.00	9.37	90-30-sf-5	4884.0	0.00	2.37	90-30-sf-5	4884.0	0.00	0.84
120-10-sf-1	2193.0	0.00	2.27	120-10-sf-1	2192.0	0.00	0.71	120-10-sf-1	2192.0	0.00	9.48
120-10-sf-2	2205.0	0.00	0.84	120-10-sf-2	2204.0	0.00	1.47	120-10-sf-2	2204.0	0.00	1.47
120-10-sf-3	2204.0	0.00	10.04	120-10-sf-3	2204.0	0.00	1.64	120-10-sf-3	2204.0	0.00	1.71
120-10-sf-4	2154.0	0.00	4.36	120-10-sf-4	2154.0	0.00	5.41	120-10-sf-4	2150.0	0.00	0.71
120-10-sf-5	2235.0	0.00	8.36	120-10-sf-5	2234.0	0.00	2.99	120-10-sf-5	2234.0	0.00	2.96
120-20-sf-1	4535.0	0.00	2.08	120-20-sf-1	4533.0	0.00	9.09	120-20-sf-1	4533.0	0.00	7.95
120-20-sf-2	4497.0	0.00	8.77	120-20-sf-2	4494.0	0.00	1.60	120-20-sf-2	4492.0	0.00	7.27
120-20-sf-3	4464.0	0.00	10.15	120-20-sf-3	4462.0	0.00	7.94	120-20-sf-3	4460.0	0.00	0.96
120-20-sf-4	-	-	-	120-20-sf-4	4385.0	0.00	8.25	120-20-sf-4	4385.0	0.00	8.46
120-20-sf-5	4447.0	0.00	67.12	120-20-sf-5	4446.0	0.00	1.09	120-20-sf-5	4446.0	0.00	1.46
120-30-sf-1	6656.0	0.00	23.77	120-30-sf-1	6655.0	0.00	1.75	120-30-sf-1	6655.0	0.00	1.04
120-30-sf-2	6727.0	0.00	56.40	120-30-sf-2	6723.0	0.00	3.77	120-30-sf-2	6722.0	0.00	1.02
120-30-sf-3	6633.0	0.00	21.89	120-30-sf-3	6633.0	0.00	1.61	120-30-sf-3	6633.0	0.00	1.10
120-30-sf-4	6639.0	0.00	23.92	120-30-sf-4	6639.0	0.00	6.56	120-30-sf-4	6636.0	0.00	1.19
120-30-sf-5	6567.0	0.00	14.01	120-30-sf-5	6565.0	0.00	5.31	120-30-sf-5	6558.0	0.00	2.19

1.3. Results on the instances with sif execution mode

Table 6 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1042.0	0.00	12.61	60-10-sif-1	1038.0	0.00	5.88	60-10-sif-1	1035.0	0.00	5.03
60-10-sif-2	-	-	-	60-10-sif-2	1062.0	0.00	3.62	60-10-sif-2	1053.0	0.00	11.62
60-10-sif-3	1076.0	0.00	6.36	60-10-sif-3	1071.0	0.00	8.44	60-10-sif-3	1068.0	0.00	3.62
60-10-sif-4	-	-	-	60-10-sif-4	1058.0	0.00	0.96	60-10-sif-4	1053.0	0.00	0.96
60-10-sif-5	1062.0	0.00	0.90	60-10-sif-5	1058.0	0.00	11.03	60-10-sif-5	1056.0	0.00	2.58
60-20-sif-1	-	-	-	60-20-sif-1	2141.0	0.00	10.00	60-20-sif-1	2136.0	0.00	9.63
60-20-sif-2	-	-	-	60-20-sif-2	2124.0	0.00	21.85	60-20-sif-2	2118.0	0.00	7.57
60-20-sif-3	-	-	-	60-20-sif-3	2027.0	0.00	6.62	60-20-sif-3	2023.0	0.00	12.52
60-20-sif-4	-	-	-	60-20-sif-4	2016.0	0.00	12.74	60-20-sif-4	2010.0	0.00	12.82
60-20-sif-5	-	-	-	60-20-sif-5	2146.0	0.00	12.33	60-20-sif-5	2138.0	0.00	3.03
60-30-sif-1	-	-	-	60-30-sif-1	3104.0	0.00	35.16	60-30-sif-1	3100.0	0.00	14.83
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	3183.0	0.00	403.44
60-30-sif-3	-	-	-	60-30-sif-3	3180.0	0.00	27.70	60-30-sif-3	3165.0	0.00	16.05
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	3161.0	0.00	109.84
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	3174.0	0.00	42.73
90-10-sif-1	-	-	-	90-10-sif-1	1624.0	0.00	18.80	90-10-sif-1	1622.0	0.00	13.65
90-10-sif-2	-	-	-	90-10-sif-2	1673.0	0.00	8.12	90-10-sif-2	1662.0	0.00	9.78
90-10-sif-3	-	-	-	90-10-sif-3	1617.0	0.00	10.01	90-10-sif-3	1612.0	0.00	0.88
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	1638.0	0.00	1.58
90-10-sif-5	1646.0	0.00	1.09	90-10-sif-5	1634.0	0.00	9.26	90-10-sif-5	1631.0	0.00	5.41
90-20-sif-1	-	-	-	90-20-sif-1	3293.0	0.00	9.98	90-20-sif-1	3279.0	0.00	19.12
90-20-sif-2	-	-	-	90-20-sif-2	3309.0	0.00	20.97	90-20-sif-2	3297.0	0.00	7.57
90-20-sif-3	-	-	-	90-20-sif-3	3306.0	0.00	24.09	90-20-sif-3	3297.0	0.00	13.20
90-20-sif-4	-	-	-	90-20-sif-4	3312.0	0.00	10.70	90-20-sif-4	3299.0	0.00	4.80
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	3261.0	0.00	26.26
90-30-sif-1	-	-	-	90-30-sif-1	4931.0	2.26	1691.78	90-30-sif-1	4909.0	0.00	1684.83
90-30-sif-2	-	-	-	90-30-sif-2	4840.0	0.00	1684.62	90-30-sif-2	4806.0	0.00	145.07
90-30-sif-3	-	-	-	90-30-sif-3	4991.0	0.00	1686.80	90-30-sif-3	4978.0	0.00	151.52
90-30-sif-4	-	-	-	90-30-sif-4	4694.0	0.00	253.58	90-30-sif-4	4679.0	0.00	711.36
90-30-sif-5	4932.0	0.00	11.81	90-30-sif-5	4900.0	0.00	101.66	90-30-sif-5	4887.0	0.00	34.80
120-10-sif-1	-	-	-	120-10-sif-1	2209.0	0.00	28.91	120-10-sif-1	2199.0	0.00	9.29
120-10-sif-2	2212.0	0.00	5.41	120-10-sif-2	2207.0	0.00	4.93	120-10-sif-2	2206.0	0.00	2.98
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	2208.0	0.00	9.61
120-10-sif-4	2167.0	0.00	1.47	120-10-sif-4	2158.0	0.00	11.93	120-10-sif-4	2155.0	0.00	8.18
120-10-sif-5	-	-	-	120-10-sif-5	2243.0	0.00	14.54	120-10-sif-5	2238.0	0.00	20.36
120-20-sif-1	-	-	-	120-20-sif-1	4543.0	0.00	32.34	120-20-sif-1	4536.0	0.00	5.19
120-20-sif-2	-	-	-	120-20-sif-2	4516.0	0.00	15.63	120-20-sif-2	4503.0	0.00	34.87
120-20-sif-3	-	-	-	120-20-sif-3	4515.0	3.02	1691.53	120-20-sif-3	4479.0	0.00	33.84
120-20-sif-4	-	-	-	120-20-sif-4	4406.0	0.00	577.97	120-20-sif-4	4394.0	0.00	75.17
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	4455.0	0.00	242.33
120-30-sif-1	-	-	-	120-30-sif-1	6695.0	2.71	1690.33	120-30-sif-1	6662.0	0.00	577.04
120-30-sif-2	-	-	-	120-30-sif-2	6747.0	0.00	1695.44	120-30-sif-2	6732.0	0.00	143.16
120-30-sif-3	-	-	-	120-30-sif-3	6652.0	0.00	382.65	120-30-sif-3	6635.0	0.00	24.52
120-30-sif-4	-	-	-	120-30-sif-4	6679.0	0.00	1685.61	120-30-sif-4	6648.0	0.00	684.75
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	6578.0	0.00	1687.62

Table 7 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1033.0	0.00	9.93	60-10-sif-1	1031.0	0.00	0.93	60-10-sif-1	1031.0	0.00	2.21
60-10-sif-2	1052.0	0.00	11.42	60-10-sif-2	1049.0	0.00	3.74	60-10-sif-2	1048.0	0.00	11.36
60-10-sif-3	1067.0	0.00	8.66	60-10-sif-3	1066.0	0.00	11.72	60-10-sif-3	1063.0	0.00	11.09
60-10-sif-4	1052.0	0.00	1.78	60-10-sif-4	1052.0	0.00	1.48	60-10-sif-4	1051.0	0.00	0.86
60-10-sif-5	1055.0	0.00	5.83	60-10-sif-5	1054.0	0.00	1.09	60-10-sif-5	1053.0	0.00	0.91
60-20-sif-1	2133.0	0.00	10.19	60-20-sif-1	2132.0	0.00	2.27	60-20-sif-1	2132.0	0.00	1.43
60-20-sif-2	2114.0	0.00	9.29	60-20-sif-2	2112.0	0.00	2.67	60-20-sif-2	2108.0	0.00	10.21
60-20-sif-3	2016.0	0.00	1.07	60-20-sif-3	2016.0	0.00	1.90	60-20-sif-3	2014.0	0.00	0.87
60-20-sif-4	2007.0	0.00	1.19	60-20-sif-4	2006.0	0.00	16.29	60-20-sif-4	2005.0	0.00	2.84
60-20-sif-5	2134.0	0.00	6.92	60-20-sif-5	2133.0	0.00	2.71	60-20-sif-5	2132.0	0.00	1.42
60-30-sif-1	3095.0	0.00	22.80	60-30-sif-1	3090.0	0.00	1.54	60-30-sif-1	3090.0	0.00	1.41
60-30-sif-2	3170.0	0.00	19.49	60-30-sif-2	3166.0	0.00	12.21	60-30-sif-2	3165.0	0.00	0.74
60-30-sif-3	3159.0	0.00	3.71	60-30-sif-3	3155.0	0.00	9.01	60-30-sif-3	3155.0	0.00	0.84
60-30-sif-4	3154.0	0.00	35.33	60-30-sif-4	3151.0	0.00	9.30	60-30-sif-4	3150.0	0.00	0.78
60-30-sif-5	3163.0	0.00	16.25	60-30-sif-5	3161.0	0.00	8.40	60-30-sif-5	3160.0	0.00	1.40
90-10-sif-1	1620.0	0.00	9.07	90-10-sif-1	1619.0	0.00	4.04	90-10-sif-1	1617.0	0.00	2.36
90-10-sif-2	1659.0	0.00	7.89	90-10-sif-2	1658.0	0.00	0.76	90-10-sif-2	1658.0	0.00	8.38
90-10-sif-3	1610.0	0.00	0.83	90-10-sif-3	1610.0	0.00	0.94	90-10-sif-3	1610.0	0.00	0.86
90-10-sif-4	1637.0	0.00	1.50	90-10-sif-4	1637.0	0.00	3.46	90-10-sif-4	1635.0	0.00	9.31
90-10-sif-5	1630.0	0.00	10.31	90-10-sif-5	1629.0	0.00	3.05	90-10-sif-5	1629.0	0.00	1.58
90-20-sif-1	3275.0	0.00	7.36	90-20-sif-1	3273.0	0.00	1.46	90-20-sif-1	3273.0	0.00	1.42
90-20-sif-2	3290.0	0.00	9.94	90-20-sif-2	3290.0	0.00	8.99	90-20-sif-2	3289.0	0.00	1.41
90-20-sif-3	3295.0	0.00	1.08	90-20-sif-3	3293.0	0.00	2.19	90-20-sif-3	3293.0	0.00	2.20
90-20-sif-4	3291.0	0.00	14.03	90-20-sif-4	3286.0	0.00	1.54	90-20-sif-4	3285.0	0.00	1.37
90-20-sif-5	3253.0	0.00	6.29	90-20-sif-5	3250.0	0.00	1.10	90-20-sif-5	3249.0	0.00	3.13
90-30-sif-1	4899.0	0.00	87.71	90-30-sif-1	4897.0	0.00	12.57	90-30-sif-1	4888.0	0.00	10.28
90-30-sif-2	4796.0	0.00	14.16	90-30-sif-2	4794.0	0.00	20.35	90-30-sif-2	4793.0	0.00	5.99
90-30-sif-3	4971.0	0.00	46.13	90-30-sif-3	4968.0	0.00	37.16	90-30-sif-3	4966.0	0.00	2.33
90-30-sif-4	4675.0	0.00	16.94	90-30-sif-4	4673.0	0.00	2.47	90-30-sif-4	4672.0	0.00	1.66
90-30-sif-5	4885.0	0.00	1.55	90-30-sif-5	4884.0	0.00	2.32	90-30-sif-5	4884.0	0.00	0.85
120-10-sif-1	2198.0	0.00	8.89	120-10-sif-1	2192.0	0.00	0.81	120-10-sif-1	2192.0	0.00	1.48
120-10-sif-2	2204.0	0.00	0.83	120-10-sif-2	2203.0	0.00	0.78	120-10-sif-2	2203.0	0.00	0.77
120-10-sif-3	2204.0	0.00	2.24	120-10-sif-3	2204.0	0.00	2.35	120-10-sif-3	2204.0	0.00	1.55
120-10-sif-4	2154.0	0.00	4.15	120-10-sif-4	2154.0	0.00	5.48	120-10-sif-4	2150.0	0.00	0.71
120-10-sif-5	2235.0	0.00	12.32	120-10-sif-5	2233.0	0.00	1.56	120-10-sif-5	2233.0	0.00	1.44
120-20-sif-1	4535.0	0.00	2.20	120-20-sif-1	4535.0	0.00	3.49	120-20-sif-1	4533.0	0.00	8.76
120-20-sif-2	4497.0	0.00	3.97	120-20-sif-2	4495.0	0.00	2.39	120-20-sif-2	4492.0	0.00	5.61
120-20-sif-3	4469.0	0.00	55.94	120-20-sif-3	4462.0	0.00	7.63	120-20-sif-3	4460.0	0.00	0.93
120-20-sif-4	4392.0	0.00	35.71	120-20-sif-4	4392.0	0.00	3.93	120-20-sif-4	4385.0	0.00	8.71
120-20-sif-5	4449.0	0.00	89.95	120-20-sif-5	4446.0	0.00	3.23	120-20-sif-5	4446.0	0.00	1.47
120-30-sif-1	6656.0	0.00	15.65	120-30-sif-1	6655.0	0.00	1.93	120-30-sif-1	6655.0	0.00	1.16
120-30-sif-2	6726.0	0.00	106.84	120-30-sif-2	6723.0	0.00	17.94	120-30-sif-2	6722.0	0.00	1.08
120-30-sif-3	6633.0	0.00	3.16	120-30-sif-3	6633.0	0.00	1.83	120-30-sif-3	6633.0	0.00	3.27
120-30-sif-4	6639.0	0.00	36.98	120-30-sif-4	6639.0	0.00	6.52	120-30-sif-4	6639.0	0.00	7.04
120-30-sif-5	6565.0	0.00	36.88	120-30-sif-5	6563.0	0.00	17.87	120-30-sif-5	6563.0	0.00	2.75

2. Detailed Results of Standard-Weight Instances on S_A under WTFRP Setting

We first report the results of Ha26 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 8 Detailed Results of Ha26 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	2301.0	0.00	1.28

2.1. Results on the instances with i execution mode

Table 9 Detailed Results of Ha26 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	2018.0	0.00	0.92
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	2295.0	0.00	10.93
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	12013.0	0.00	32.95
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	11252.0	0.00	56.86
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 10 Detailed Results of Ha26 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1881.0	0.00	9.72	60-10-i-1	1879.0	0.00	0.81
60-10-i-2	2018.0	0.00	8.56	60-10-i-2	2012.0	0.00	0.98	60-10-i-2	2011.0	0.00	0.83
60-10-i-3	2289.0	0.00	0.88	60-10-i-3	2289.0	0.00	9.41	60-10-i-3	2289.0	0.00	10.55
60-10-i-4	-	-	-	60-10-i-4	2034.0	0.00	0.96	60-10-i-4	2034.0	0.00	0.93
60-10-i-5	1927.0	0.00	0.94	60-10-i-5	1927.0	0.00	1.70	60-10-i-5	1925.0	0.00	0.86
60-20-i-1	3772.0	0.00	0.84	60-20-i-1	3770.0	0.00	1.36	60-20-i-1	3770.0	0.00	1.23
60-20-i-2	3953.0	0.00	9.28	60-20-i-2	3948.0	0.00	7.60	60-20-i-2	3946.0	0.00	0.66
60-20-i-3	3200.0	0.00	8.36	60-20-i-3	3197.0	0.00	2.83	60-20-i-3	3196.0	0.00	0.75
60-20-i-4	-	-	-	60-20-i-4	3848.0	0.00	0.79	60-20-i-4	3844.0	0.00	0.76
60-20-i-5	-	-	-	60-20-i-5	3658.0	0.00	1.26	60-20-i-5	3658.0	0.00	0.63
60-30-i-1	6739.0	0.00	8.48	60-30-i-1	6730.0	0.00	0.82	60-30-i-1	6724.0	0.00	0.67
60-30-i-2	-	-	-	60-30-i-2	5798.0	0.00	10.08	60-30-i-2	5786.0	0.00	9.09
60-30-i-3	-	-	-	60-30-i-3	5338.0	0.00	9.21	60-30-i-3	5338.0	0.00	1.29
60-30-i-4	-	-	-	60-30-i-4	5536.0	0.00	10.63	60-30-i-4	5533.0	0.00	3.76
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	5328.0	0.00	1.35
90-10-i-1	-	-	-	90-10-i-1	2950.0	0.00	1.87	90-10-i-1	2950.0	0.00	7.33
90-10-i-2	-	-	-	90-10-i-2	3022.0	0.00	0.73	90-10-i-2	3022.0	0.00	0.76
90-10-i-3	2946.0	0.00	0.79	90-10-i-3	2946.0	0.00	9.61	90-10-i-3	2944.0	0.00	0.75
90-10-i-4	-	-	-	90-10-i-4	3188.0	0.00	9.32	90-10-i-4	3186.0	0.00	0.75
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	2641.0	0.00	9.47
90-20-i-1	-	-	-	90-20-i-1	6565.0	0.00	1.35	90-20-i-1	6565.0	0.00	1.27
90-20-i-2	-	-	-	90-20-i-2	6874.0	0.00	7.49	90-20-i-2	6873.0	0.00	6.66
90-20-i-3	6187.0	0.00	6.85	90-20-i-3	6178.0	0.00	7.88	90-20-i-3	6175.0	0.00	0.65
90-20-i-4	5309.0	0.00	3.29	90-20-i-4	5302.0	0.00	4.74	90-20-i-4	5301.0	0.00	7.22
90-20-i-5	-	-	-	90-20-i-5	6029.0	0.00	6.99	90-20-i-5	6025.0	0.00	4.54
90-30-i-1	9636.0	0.00	2.51	90-30-i-1	9625.0	0.00	0.78	90-30-i-1	9625.0	0.00	0.71
90-30-i-2	-	-	-	90-30-i-2	8587.0	0.00	4.25	90-30-i-2	8581.0	0.00	1.47
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	8719.0	0.00	0.78
90-30-i-4	9028.0	0.00	6.69	90-30-i-4	9026.0	0.00	1.88	90-30-i-4	9026.0	0.00	0.69
90-30-i-5	8044.0	0.00	1.39	90-30-i-5	8040.0	0.00	2.73	90-30-i-5	8040.0	0.00	1.38
120-10-i-1	3886.0	0.00	0.79	120-10-i-1	3886.0	0.00	9.65	120-10-i-1	3886.0	0.00	0.71
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	3571.0	0.00	0.70
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	3440.0	0.00	0.72
120-10-i-4	-	-	-	120-10-i-4	4352.0	0.00	0.70	120-10-i-4	4352.0	0.00	0.74
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	3857.0	0.00	9.66
120-20-i-1	8143.0	0.00	1.37	120-20-i-1	8143.0	0.00	1.29	120-20-i-1	8143.0	0.00	1.30
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	9375.0	0.00	0.77
120-20-i-3	9751.0	0.00	10.82	120-20-i-3	9742.0	0.00	0.88	120-20-i-3	9742.0	0.00	0.72
120-20-i-4	7794.0	0.00	11.74	120-20-i-4	7792.0	0.00	2.08	120-20-i-4	7792.0	0.00	2.45
120-20-i-5	7532.0	0.00	10.11	120-20-i-5	7532.0	0.00	26.82	120-20-i-5	7532.0	0.00	9.12
120-30-i-1	13088.0	0.00	86.14	120-30-i-1	13080.0	0.00	12.12	120-30-i-1	13080.0	0.00	1.78
120-30-i-2	12002.0	0.00	9.09	120-30-i-2	12002.0	0.00	0.91	120-30-i-2	12002.0	0.00	1.46
120-30-i-3	11246.0	0.00	8.93	120-30-i-3	11246.0	0.00	1.79	120-30-i-3	11246.0	0.00	1.37
120-30-i-4	11496.0	0.00	2.61	120-30-i-4	11496.0	0.00	2.08	120-30-i-4	11490.0	0.00	1.42
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	13444.0	0.00	2.63

2.2. Results on the instances with sf execution mode

Table 11 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1884.0	0.00	10.08	60-10-sf-1	1867.0	0.00	6.04	60-10-sf-1	1863.0	0.00	3.34
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1997.0	0.00	8.84
60-10-sf-3	2281.0	0.00	9.59	60-10-sf-3	2267.0	0.00	2.46	60-10-sf-3	2263.0	0.00	8.70
60-10-sf-4	2034.0	0.00	1.00	60-10-sf-4	2029.0	0.00	10.94	60-10-sf-4	2011.0	0.00	0.96
60-10-sf-5	1915.0	0.00	1.99	60-10-sf-5	1912.0	0.00	1.86	60-10-sf-5	1908.0	0.00	4.13
60-20-sf-1	-	-	-	60-20-sf-1	3756.0	0.00	10.04	60-20-sf-1	3744.0	0.00	3.81
60-20-sf-2	-	-	-	60-20-sf-2	3944.0	0.00	11.70	60-20-sf-2	3931.0	0.00	7.27
60-20-sf-3	-	-	-	60-20-sf-3	3200.0	0.00	6.34	60-20-sf-3	3187.0	0.00	10.82
60-20-sf-4	-	-	-	60-20-sf-4	3829.0	0.00	10.34	60-20-sf-4	3812.0	0.00	8.99
60-20-sf-5	-	-	-	60-20-sf-5	3648.0	0.00	9.48	60-20-sf-5	3634.0	0.00	8.41
60-30-sf-1	-	-	-	60-30-sf-1	6677.0	0.00	16.16	60-30-sf-1	6656.0	0.00	11.03
60-30-sf-2	-	-	-	60-30-sf-2	5775.0	0.00	568.06	60-30-sf-2	5734.0	0.00	68.97
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	5298.0	0.00	12.55
60-30-sf-4	-	-	-	60-30-sf-4	5524.0	0.00	232.89	60-30-sf-4	5495.0	0.00	83.79
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	5291.0	0.00	29.12
90-10-sf-1	-	-	-	90-10-sf-1	2926.0	0.00	13.36	90-10-sf-1	2919.0	0.00	8.15
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	2996.0	0.00	1.92
90-10-sf-3	-	-	-	90-10-sf-3	2919.0	0.00	1.63	90-10-sf-3	2916.0	0.00	0.84
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	2633.0	0.00	13.13	90-10-sf-5	2621.0	0.00	11.03	90-10-sf-5	2615.0	0.00	8.47
90-20-sf-1	-	-	-	90-20-sf-1	6522.0	0.00	13.32	90-20-sf-1	6496.0	0.00	3.03
90-20-sf-2	-	-	-	90-20-sf-2	6804.0	0.00	30.29	90-20-sf-2	6783.0	0.00	25.34
90-20-sf-3	-	-	-	90-20-sf-3	6133.0	0.00	16.20	90-20-sf-3	6109.0	0.00	13.04
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	5269.0	0.00	2.35
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	5970.0	0.00	12.65
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	9560.0	0.00	799.18
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	8537.0	0.00	193.99
90-30-sf-3	-	-	-	90-30-sf-3	8668.0	0.00	662.05	90-30-sf-3	8645.0	0.00	148.53
90-30-sf-4	-	-	-	90-30-sf-4	8999.0	0.00	772.24	90-30-sf-4	8946.0	0.00	208.51
90-30-sf-5	-	-	-	90-30-sf-5	7986.0	0.00	80.31	90-30-sf-5	7964.0	0.00	2.42
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	3853.0	0.00	10.17
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	3553.0	0.00	8.71
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	3411.0	0.00	3.89
120-10-sf-4	4345.0	0.00	9.02	120-10-sf-4	4307.0	0.00	3.82	120-10-sf-4	4307.0	0.00	9.73
120-10-sf-5	-	-	-	120-10-sf-5	3846.0	0.00	10.05	120-10-sf-5	3823.0	0.00	10.62
120-20-sf-1	8085.0	0.00	67.16	120-20-sf-1	8058.0	0.00	15.80	120-20-sf-1	8051.0	0.00	3.14
120-20-sf-2	-	-	-	120-20-sf-2	9302.0	0.00	13.60	120-20-sf-2	9277.0	0.00	13.17
120-20-sf-3	-	-	-	120-20-sf-3	9692.0	0.00	1399.40	120-20-sf-3	9625.0	0.00	52.50
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	13009.0	2.91	1689.86	120-30-sf-1	12963.0	0.00	51.35
120-30-sf-2	-	-	-	120-30-sf-2	11934.0	0.00	1161.05	120-30-sf-2	11896.0	0.00	45.52
120-30-sf-3	-	-	-	120-30-sf-3	11169.0	0.00	115.06	120-30-sf-3	11156.0	0.00	34.26
120-30-sf-4	-	-	-	120-30-sf-4	11412.0	0.00	1686.28	120-30-sf-4	11384.0	0.00	1684.74
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 12 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1859.0	0.00	0.94	60-10-sf-1	1859.0	0.00	0.99	60-10-sf-1	1859.0	0.00	1.52
60-10-sf-2	1991.0	0.00	6.91	60-10-sf-2	1991.0	0.00	6.88	60-10-sf-2	1989.0	0.00	10.47
60-10-sf-3	2260.0	0.00	1.69	60-10-sf-3	2251.0	0.00	10.47	60-10-sf-3	2251.0	0.00	11.08
60-10-sf-4	2009.0	0.00	1.78	60-10-sf-4	2009.0	0.00	1.49	60-10-sf-4	2007.0	0.00	0.86
60-10-sf-5	1903.0	0.00	1.84	60-10-sf-5	1902.0	0.00	1.06	60-10-sf-5	1900.0	0.00	0.87
60-20-sf-1	3738.0	0.00	0.94	60-20-sf-1	3738.0	0.00	0.85	60-20-sf-1	3738.0	0.00	1.56
60-20-sf-2	3918.0	0.00	10.36	60-20-sf-2	3902.0	0.00	9.68	60-20-sf-2	3900.0	0.00	9.89
60-20-sf-3	3178.0	0.00	1.02	60-20-sf-3	3178.0	0.00	3.29	60-20-sf-3	3175.0	0.00	0.81
60-20-sf-4	3807.0	0.00	9.18	60-20-sf-4	3805.0	0.00	9.21	60-20-sf-4	3803.0	0.00	2.16
60-20-sf-5	3629.0	0.00	0.76	60-20-sf-5	3628.0	0.00	4.37	60-20-sf-5	3626.0	0.00	1.35
60-30-sf-1	6647.0	0.00	9.98	60-30-sf-1	6642.0	0.00	3.67	60-30-sf-1	6642.0	0.00	5.14
60-30-sf-2	5710.0	0.00	52.65	60-30-sf-2	5705.0	0.00	10.53	60-30-sf-2	5704.0	0.00	0.83
60-30-sf-3	5283.0	0.00	6.74	60-30-sf-3	5277.0	0.00	1.94	60-30-sf-3	5277.0	0.00	0.81
60-30-sf-4	5480.0	0.00	18.04	60-30-sf-4	5473.0	0.00	11.12	60-30-sf-4	5472.0	0.00	8.64
60-30-sf-5	5278.0	0.00	68.17	60-30-sf-5	5274.0	0.00	5.40	60-30-sf-5	5273.0	0.00	1.41
90-10-sf-1	2919.0	0.00	7.44	90-10-sf-1	2908.0	0.00	8.93	90-10-sf-1	2908.0	0.00	3.22
90-10-sf-2	2990.0	0.00	7.88	90-10-sf-2	2988.0	0.00	0.75	90-10-sf-2	2988.0	0.00	8.31
90-10-sf-3	2915.0	0.00	0.86	90-10-sf-3	2915.0	0.00	1.74	90-10-sf-3	2915.0	0.00	0.86
90-10-sf-4	3162.0	0.00	8.95	90-10-sf-4	3153.0	0.00	1.54	90-10-sf-4	3153.0	0.00	9.28
90-10-sf-5	2612.0	0.00	9.10	90-10-sf-5	2612.0	0.00	8.90	90-10-sf-5	2612.0	0.00	1.57
90-20-sf-1	6490.0	0.00	6.53	90-20-sf-1	6487.0	0.00	1.95	90-20-sf-1	6487.0	0.00	7.61
90-20-sf-2	6778.0	0.00	8.90	90-20-sf-2	6778.0	0.00	8.66	90-20-sf-2	6775.0	0.00	1.39
90-20-sf-3	6107.0	0.00	8.00	90-20-sf-3	6107.0	0.00	6.66	90-20-sf-3	6107.0	0.00	7.28
90-20-sf-4	5260.0	0.00	9.43	90-20-sf-4	5252.0	0.00	2.26	90-20-sf-4	5250.0	0.00	0.78
90-20-sf-5	5961.0	0.00	9.52	90-20-sf-5	5954.0	0.00	1.52	90-20-sf-5	5952.0	0.00	1.51
90-30-sf-1	9533.0	0.00	51.27	90-30-sf-1	9500.0	0.00	10.11	90-30-sf-1	9496.0	0.00	9.96
90-30-sf-2	8501.0	0.00	14.10	90-30-sf-2	8493.0	0.00	9.40	90-30-sf-2	8482.0	0.00	0.93
90-30-sf-3	8636.0	0.00	27.01	90-30-sf-3	8633.0	0.00	32.52	90-30-sf-3	8628.0	0.00	2.30
90-30-sf-4	8933.0	0.00	11.84	90-30-sf-4	8931.0	0.00	1.88	90-30-sf-4	8929.0	0.00	0.90
90-30-sf-5	7961.0	0.00	5.53	90-30-sf-5	7959.0	0.00	3.34	90-30-sf-5	7959.0	0.00	0.77
120-10-sf-1	3851.0	0.00	2.27	120-10-sf-1	3850.0	0.00	0.71	120-10-sf-1	3850.0	0.00	9.25
120-10-sf-2	3553.0	0.00	0.78	120-10-sf-2	3551.0	0.00	1.42	120-10-sf-2	3551.0	0.00	1.43
120-10-sf-3	3407.0	0.00	9.07	120-10-sf-3	3407.0	0.00	2.23	120-10-sf-3	3407.0	0.00	1.56
120-10-sf-4	4305.0	0.00	5.95	120-10-sf-4	4305.0	0.00	4.10	120-10-sf-4	4289.0	0.00	0.70
120-10-sf-5	3822.0	0.00	8.37	120-10-sf-5	3820.0	0.00	2.80	120-10-sf-5	3820.0	0.00	3.23
120-20-sf-1	8049.0	0.00	2.65	120-20-sf-1	8043.0	0.00	2.53	120-20-sf-1	8043.0	0.00	9.10
120-20-sf-2	9260.0	0.00	4.23	120-20-sf-2	9251.0	0.00	2.22	120-20-sf-2	9246.0	0.00	6.06
120-20-sf-3	9598.0	0.00	16.64	120-20-sf-3	9591.0	0.00	12.64	120-20-sf-3	9587.0	0.00	0.88
120-20-sf-4	-	-	-	120-20-sf-4	7695.0	0.00	8.52	120-20-sf-4	7695.0	0.00	8.29
120-20-sf-5	7457.0	0.00	3.07	120-20-sf-5	7455.0	0.00	0.92	120-20-sf-5	7455.0	0.00	1.38
120-30-sf-1	12955.0	0.00	8.35	120-30-sf-1	12954.0	0.00	1.10	120-30-sf-1	12954.0	0.00	1.48
120-30-sf-2	11889.0	0.00	160.31	120-30-sf-2	11881.0	0.00	4.70	120-30-sf-2	11879.0	0.00	102.28
120-30-sf-3	11152.0	0.00	1.60	120-30-sf-3	11152.0	0.00	1.15	120-30-sf-3	11152.0	0.00	3.27
120-30-sf-4	11366.0	0.00	49.91	120-30-sf-4	11366.0	0.00	12.35	120-30-sf-4	11357.0	0.00	1.14
120-30-sf-5	13291.0	0.00	24.25	120-30-sf-5	13287.0	0.00	3.56	120-30-sf-5	13266.0	0.00	1.94

2.3. Results on the instances with sif execution mode

Table 13 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1880.0	0.00	1.39	60-10-sif-1	1872.0	0.00	7.42	60-10-sif-1	1866.0	0.00	8.53
60-10-sif-2	-	-	-	60-10-sif-2	2018.0	0.00	11.38	60-10-sif-2	2003.0	0.00	10.57
60-10-sif-3	2287.0	0.00	2.76	60-10-sif-3	2271.0	0.00	9.91	60-10-sif-3	2265.0	0.00	9.15
60-10-sif-4	-	-	-	60-10-sif-4	2025.0	0.00	0.95	60-10-sif-4	2011.0	0.00	0.96
60-10-sif-5	1917.0	0.00	0.96	60-10-sif-5	1910.0	0.00	4.01	60-10-sif-5	1906.0	0.00	1.83
60-20-sif-1	-	-	-	60-20-sif-1	3753.0	0.00	10.14	60-20-sif-1	3745.0	0.00	1.24
60-20-sif-2	-	-	-	60-20-sif-2	3932.0	0.00	1.32	60-20-sif-2	3922.0	0.00	2.55
60-20-sif-3	-	-	-	60-20-sif-3	3202.0	0.00	11.81	60-20-sif-3	3189.0	0.00	5.19
60-20-sif-4	-	-	-	60-20-sif-4	3824.0	0.00	4.40	60-20-sif-4	3811.0	0.00	3.27
60-20-sif-5	-	-	-	60-20-sif-5	3652.0	0.00	13.43	60-20-sif-5	3637.0	0.00	8.60
60-30-sif-1	-	-	-	60-30-sif-1	6670.0	0.00	17.95	60-30-sif-1	6659.0	0.00	9.62
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	5741.0	0.00	314.94
60-30-sif-3	-	-	-	60-30-sif-3	5324.0	0.00	19.44	60-30-sif-3	5293.0	0.00	13.36
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	5492.0	0.00	13.05
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	5296.0	0.00	33.00
90-10-sif-1	-	-	-	90-10-sif-1	2925.0	0.00	32.19	90-10-sif-1	2919.0	0.00	10.77
90-10-sif-2	-	-	-	90-10-sif-2	3017.0	0.00	7.94	90-10-sif-2	2996.0	0.00	8.58
90-10-sif-3	-	-	-	90-10-sif-3	2930.0	0.00	9.81	90-10-sif-3	2917.0	0.00	0.88
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	3162.0	0.00	0.90
90-10-sif-5	2647.0	0.00	1.10	90-10-sif-5	2621.0	0.00	13.57	90-10-sif-5	2615.0	0.00	9.66
90-20-sif-1	-	-	-	90-20-sif-1	6536.0	0.00	12.80	90-20-sif-1	6496.0	0.00	10.67
90-20-sif-2	-	-	-	90-20-sif-2	6816.0	0.00	4.97	90-20-sif-2	6797.0	0.00	22.68
90-20-sif-3	-	-	-	90-20-sif-3	6132.0	0.00	5.21	90-20-sif-3	6111.0	0.00	14.65
90-20-sif-4	-	-	-	90-20-sif-4	5292.0	0.00	11.59	90-20-sif-4	5269.0	0.00	2.51
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	5974.0	0.00	40.22
90-30-sif-1	-	-	-	90-30-sif-1	9588.0	0.00	316.90	90-30-sif-1	9544.0	0.00	1682.57
90-30-sif-2	-	-	-	90-30-sif-2	8591.0	0.00	152.03	90-30-sif-2	8515.0	0.00	186.84
90-30-sif-3	-	-	-	90-30-sif-3	8673.0	0.00	202.58	90-30-sif-3	8648.0	0.00	111.62
90-30-sif-4	-	-	-	90-30-sif-4	8981.0	0.00	369.91	90-30-sif-4	8942.0	0.00	294.50
90-30-sif-5	8049.0	0.00	9.22	90-30-sif-5	7989.0	0.00	80.27	90-30-sif-5	7964.0	0.00	21.76
120-10-sif-1	-	-	-	120-10-sif-1	3881.0	0.00	24.15	120-10-sif-1	3867.0	0.00	9.95
120-10-sif-2	3563.0	0.00	1.12	120-10-sif-2	3555.0	0.00	0.86	120-10-sif-2	3554.0	0.00	3.50
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	3413.0	0.00	10.87
120-10-sif-4	4345.0	0.00	9.58	120-10-sif-4	4313.0	0.00	10.11	120-10-sif-4	4307.0	0.00	9.22
120-10-sif-5	-	-	-	120-10-sif-5	3839.0	0.00	10.63	120-10-sif-5	3827.0	0.00	13.66
120-20-sif-1	-	-	-	120-20-sif-1	8062.0	0.00	26.17	120-20-sif-1	8051.0	0.00	14.54
120-20-sif-2	-	-	-	120-20-sif-2	9309.0	0.00	18.18	120-20-sif-2	9273.0	0.00	13.38
120-20-sif-3	-	-	-	120-20-sif-3	9723.0	0.00	1207.90	120-20-sif-3	9638.0	0.00	53.75
120-20-sif-4	-	-	-	120-20-sif-4	7739.0	0.00	467.72	120-20-sif-4	7719.0	0.00	46.20
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	7470.0	0.00	264.31
120-30-sif-1	-	-	-	120-30-sif-1	13038.0	2.17	1684.93	120-30-sif-1	12967.0	0.00	326.72
120-30-sif-2	-	-	-	120-30-sif-2	11922.0	0.00	582.50	120-30-sif-2	11896.0	0.00	378.82
120-30-sif-3	-	-	-	120-30-sif-3	11191.0	0.00	265.65	120-30-sif-3	11156.0	0.00	21.93
120-30-sif-4	-	-	-	120-30-sif-4	11440.0	0.00	1683.71	120-30-sif-4	11382.0	0.00	1684.61
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	13314.0	0.00	1333.01

Table 14 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1862.0	0.00	8.05	60-10-sif-1	1859.0	0.00	0.96	60-10-sif-1	1859.0	0.00	1.52
60-10-sif-2	1999.0	0.00	11.78	60-10-sif-2	1991.0	0.00	3.82	60-10-sif-2	1989.0	0.00	11.44
60-10-sif-3	2261.0	0.00	8.01	60-10-sif-3	2260.0	0.00	11.54	60-10-sif-3	2251.0	0.00	11.22
60-10-sif-4	2009.0	0.00	1.78	60-10-sif-4	2009.0	0.00	1.49	60-10-sif-4	2007.0	0.00	0.86
60-10-sif-5	1903.0	0.00	1.89	60-10-sif-5	1902.0	0.00	1.08	60-10-sif-5	1900.0	0.00	0.86
60-20-sif-1	3740.0	0.00	10.29	60-20-sif-1	3738.0	0.00	0.95	60-20-sif-1	3738.0	0.00	1.46
60-20-sif-2	3916.0	0.00	3.81	60-20-sif-2	3911.0	0.00	1.87	60-20-sif-2	3900.0	0.00	8.62
60-20-sif-3	3178.0	0.00	1.13	60-20-sif-3	3178.0	0.00	2.70	60-20-sif-3	3175.0	0.00	0.85
60-20-sif-4	3807.0	0.00	1.82	60-20-sif-4	3805.0	0.00	9.25	60-20-sif-4	3803.0	0.00	2.16
60-20-sif-5	3629.0	0.00	8.26	60-20-sif-5	3628.0	0.00	3.85	60-20-sif-5	3626.0	0.00	1.28
60-30-sif-1	6647.0	0.00	10.49	60-30-sif-1	6636.0	0.00	1.58	60-30-sif-1	6636.0	0.00	1.40
60-30-sif-2	5712.0	0.00	56.96	60-30-sif-2	5705.0	0.00	16.34	60-30-sif-2	5704.0	0.00	0.74
60-30-sif-3	5283.0	0.00	13.27	60-30-sif-3	5277.0	0.00	1.64	60-30-sif-3	5277.0	0.00	2.05
60-30-sif-4	5479.0	0.00	22.26	60-30-sif-4	5473.0	0.00	10.97	60-30-sif-4	5472.0	0.00	1.71
60-30-sif-5	5278.0	0.00	14.98	60-30-sif-5	5274.0	0.00	9.12	60-30-sif-5	5273.0	0.00	1.38
90-10-sif-1	2916.0	0.00	7.86	90-10-sif-1	2914.0	0.00	4.05	90-10-sif-1	2907.0	0.00	9.40
90-10-sif-2	2990.0	0.00	7.58	90-10-sif-2	2988.0	0.00	8.51	90-10-sif-2	2988.0	0.00	8.35
90-10-sif-3	2915.0	0.00	0.82	90-10-sif-3	2915.0	0.00	1.74	90-10-sif-3	2915.0	0.00	0.87
90-10-sif-4	3160.0	0.00	2.90	90-10-sif-4	3159.0	0.00	4.39	90-10-sif-4	3153.0	0.00	10.02
90-10-sif-5	2614.0	0.00	7.97	90-10-sif-5	2612.0	0.00	2.30	90-10-sif-5	2612.0	0.00	1.58
90-20-sif-1	6490.0	0.00	10.37	90-20-sif-1	6487.0	0.00	1.48	90-20-sif-1	6487.0	0.00	0.87
90-20-sif-2	6778.0	0.00	10.34	90-20-sif-2	6778.0	0.00	8.53	90-20-sif-2	6775.0	0.00	1.43
90-20-sif-3	6107.0	0.00	1.05	90-20-sif-3	6101.0	0.00	5.26	90-20-sif-3	6101.0	0.00	2.75
90-20-sif-4	5259.0	0.00	10.08	90-20-sif-4	5252.0	0.00	1.74	90-20-sif-4	5250.0	0.00	1.96
90-20-sif-5	5959.0	0.00	1.66	90-20-sif-5	5954.0	0.00	8.62	90-20-sif-5	5952.0	0.00	2.25
90-30-sif-1	9527.0	0.00	50.83	90-30-sif-1	9521.0	0.00	3.41	90-30-sif-1	9496.0	0.00	10.66
90-30-sif-2	8497.0	0.00	6.43	90-30-sif-2	8493.0	0.00	15.38	90-30-sif-2	8491.0	0.00	5.28
90-30-sif-3	8638.0	0.00	12.12	90-30-sif-3	8633.0	0.00	29.11	90-30-sif-3	8628.0	0.00	2.48
90-30-sif-4	8935.0	0.00	5.18	90-30-sif-4	8931.0	0.00	1.30	90-30-sif-4	8929.0	0.00	0.96
90-30-sif-5	7961.0	0.00	2.07	90-30-sif-5	7959.0	0.00	1.71	90-30-sif-5	7959.0	0.00	0.88
120-10-sif-1	3866.0	0.00	8.47	120-10-sif-1	3850.0	0.00	0.79	120-10-sif-1	3850.0	0.00	1.45
120-10-sif-2	3551.0	0.00	0.82	120-10-sif-2	3549.0	0.00	0.78	120-10-sif-2	3549.0	0.00	0.77
120-10-sif-3	3407.0	0.00	6.49	120-10-sif-3	3407.0	0.00	0.85	120-10-sif-3	3407.0	0.00	0.83
120-10-sif-4	4305.0	0.00	7.94	120-10-sif-4	4305.0	0.00	5.48	120-10-sif-4	4289.0	0.00	0.72
120-10-sif-5	3821.0	0.00	10.18	120-10-sif-5	3818.0	0.00	0.83	120-10-sif-5	3818.0	0.00	8.53
120-20-sif-1	8049.0	0.00	1.86	120-20-sif-1	8049.0	0.00	2.58	120-20-sif-1	8043.0	0.00	8.39
120-20-sif-2	9260.0	0.00	4.36	120-20-sif-2	9255.0	0.00	1.60	120-20-sif-2	9246.0	0.00	7.54
120-20-sif-3	9611.0	0.00	49.00	120-20-sif-3	9591.0	0.00	5.22	120-20-sif-3	9587.0	0.00	1.49
120-20-sif-4	7716.0	0.00	5.37	120-20-sif-4	7716.0	0.00	4.56	120-20-sif-4	7695.0	0.00	8.39
120-20-sif-5	7460.0	0.00	95.92	120-20-sif-5	7455.0	0.00	1.88	120-20-sif-5	7455.0	0.00	1.42
120-30-sif-1	12955.0	0.00	15.81	120-30-sif-1	12954.0	0.00	1.31	120-30-sif-1	12954.0	0.00	1.14
120-30-sif-2	11886.0	0.00	58.89	120-30-sif-2	11881.0	0.00	5.07	120-30-sif-2	11879.0	0.00	59.00
120-30-sif-3	11152.0	0.00	102.17	120-30-sif-3	11152.0	0.00	1.74	120-30-sif-3	11152.0	0.00	3.36
120-30-sif-4	11366.0	0.00	51.08	120-30-sif-4	11366.0	0.00	25.65	120-30-sif-4	11366.0	0.00	4.97
120-30-sif-5	13285.0	0.00	83.12	120-30-sif-5	13281.0	0.00	5.98	120-30-sif-5	13281.0	0.00	3.32